

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5421

5 Credits: 4

Program core

Source-grounded

Unit Operations of Food Industry

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5421 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5421.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Food Processing Technology
Course code	5421
Course title	Unit Operations of Food Industry
Semester	5 Credits: 4
Course category	Program core
Periods per week	4(L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	17 Understanding processing and different types of evaporators Describe the principles and mechanisms of freezing	See official syllabus
CO2	17 Understanding and freezing systems Describe the principles, working mechanisms, and applications of various food processing equipment,	See official syllabus
CO3	including hammer mills, ball mills, attrition mills, 13 Understanding and colloidal mills, in relation to energy laws and their effects on food. Describe the theory, importance, equipment, mechanisms, and applications of filtration,	See official syllabus
CO4	including plate and frame filters, rotary filters, and 13 Understanding the use of filter aids.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	of evaporators	3	As specified
Module 2	Describe the principles and mechanisms of freezing and freezing systems	4	As specified
Module 3	Describe the principles, working mechanisms, and applications of various	3	As specified
Module 4	applications of filtration, including plate and frame filters, rotary filters,	3	As specified

MODULE 1

of evaporators

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of evaporators
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of

Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of is an official topic in Module 1 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the principle of evaporation 6 understanding explain the working principle and mechanism of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of definition/meaning . steps, application . Technical terms English- .

▼ 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect

7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect is an official topic in Module 1 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 understanding natural, forced circulation & falling thin film evaporator- describe single effect and multiple effect should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect definition/meaning. definition, steps, application, Technical terms English- Malayalam.

▼ 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators.

4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. is an official topic in Module 1 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators. ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English-ഫലപ്രസാദം ഫലപ്രസാദം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Describe the principles and mechanisms of freezing and freezing systems

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the principles and mechanisms of freezing and freezing systems
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Describe the Principle of freezing 7 Understanding List the factors influencing freezing time

Describe the Principle of freezing 7 Understanding List the factors influencing freezing time is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the Principle of freezing 7 Understanding List the factors influencing freezing time using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the principle of freezing 7 understanding list the factors influencing freezing time should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Describe the Principle of freezing 7 Understanding List the factors influencing freezing time means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Describe the Principle of freezing 7 Understanding List the factors influencing freezing time definition/meaning . steps, application . Technical terms English- .

▼ 3 Remembering Describe the direct freezing systems

3 Remembering Describe the direct freezing systems is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering Describe the direct freezing systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering describe the direct freezing systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering Describe the direct freezing systems means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Remembering Describe the direct freezing systems
 definition/meaning
 steps, application
 Technical terms English-

▼ 3 Understanding Describe the Indirect freezing systems

3 Understanding Describe the Indirect freezing systems is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Describe the Indirect freezing systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding describe the indirect freezing systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Describe the Indirect freezing systems means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding Describe the Indirect freezing systems
 definition/meaning
 , steps, application
 Technical terms English-

▼ 3 Understanding factors influencing freezing time, Freezing systems; Direct contact systems (air blast, immersion) & indirect contact systems

3 Understanding factors influencing freezing time, Freezing systems; Direct contact systems (air blast, immersion) & indirect contact systems is an official topic in Module 2 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding factors influencing freezing time, Freezing systems; Direct contact systems (air blast, immersion) & indirect contact systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding factors influencing freezing time, freezing systems; direct contact systems (air blast, immersion) & indirect contact systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding factors influencing freezing time, Freezing systems; Direct contact systems (air blast, immersion) & indirect contact systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Understanding factors influencing freezing time, Freezing systems; Direct contact systems (air blast, immersion) & indirect contact systems definition/meaning. steps, application. Technical terms English- Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Describe the principles, working mechanisms, and applications of various

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the principles, working mechanisms, and applications of various
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ laws 5 Understanding Discuss the equipment of size reduction in solid

laws 5 Understanding Discuss the equipment of size reduction in solid is an official topic in Module 3 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify laws 5 Understanding Discuss the equipment of size reduction in solid using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, laws 5 understanding discuss the equipment of size reduction in solid should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What laws 5 Understanding Discuss the equipment of size reduction in solid means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- laws 5 Understanding Discuss the equipment of size reduction in solid definition/meaning . steps, application . Technical terms English- .

▼ food 5 Understanding Discuss the equipment of size reduction in liquid

food 5 Understanding Discuss the equipment of size reduction in liquid is an official topic in Module 3 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify food 5 Understanding Discuss the equipment of size reduction in liquid using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, food 5 understanding discuss the equipment of size reduction in liquid should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What food 5 Understanding Discuss the equipment of size reduction in liquid means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- food 5 Understanding Discuss the equipment of size reduction in liquid definition/meaning, steps, application Technical terms English-

▼ 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and

3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and is an official topic in Module 3 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 food understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill describe the theory, importance, equipment, mechanisms, and should be discussed with attention to

safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and definition/meaning. steps, application. Technical terms English- .

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the

Malayalam support notes.

MODULE 4

applications of filtration, including plate and frame filters, rotary filters,

This module is presented from the official syllabus scope for Unit Operations of Food Industry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: applications of filtration, including plate and frame filters, rotary filters,
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ importance of filtration

importance of filtration is an official topic in Module 4 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify importance of filtration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, importance of filtration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What importance of filtration means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓർമ്മിക്കുക ഫിലട്രേഷൻ ഫിലട്രേഷൻ ഫിലട്രേഷൻ ഫിലട്രേഷൻ definition/meaning ഫിലട്രേഷൻ ഫിലട്രേഷൻ. ഫിലട്രേഷൻ ഫിലട്രേഷൻ ഫിലട്രേഷൻ, steps, application ഫിലട്രേഷൻ ഫിലട്രേഷൻ ഫിലട്രേഷൻ. Technical terms English-ഓർമ്മിക്കുക ഫിലട്രേഷൻ ഫിലട്രേഷൻ.

▼ Discuss about the filtration equipment

Discuss about the filtration equipment is an official topic in Module 4 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss about the filtration equipment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss about the filtration equipment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss about the filtration equipment means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Discuss about the filtration equipment
 ഫിലട്രേഷൻ യൂണിറ്റ് definition/meaning എഴുതുക. ഫിലട്രേഷൻ യൂണിറ്റ്
 ഫലപ്രസാദം, steps, application എഴുതുക ഫിലട്രേഷൻ യൂണിറ്റ്. Technical terms
 English-ൽ എഴുതുക ഫിലട്രേഷൻ യൂണിറ്റ്.

▼ **Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth - Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 <https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html> 2 <https://www.sciencedirect.com/science/article/abs/pii/S0032591075850042> 3 <https://savree.com/en/encyclopedia/ball-mill> 4 <https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html>**

Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th R5 Edition, Vol. I & II, Butterworth - Heinman / Elsevier, 2004 Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1

<https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html> 2

<https://www.sciencedirect.com/science/article/abs/pii/S0032591075850042>

3 <https://savree.com/en/encyclopedia/ball-mill> 4

<https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html> is an official topic in Module 4 of Unit Operations of Food Industry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. "Coulson & Richardson's Chemical Engineering", 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., " Principles of Unit Operations", 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 <https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html> 2 <https://www.sciencedirect.com/science/article/abs/pii/S0032591075850042> 3 <https://savree.com/en/encyclopedia/ball-mill> 4 <https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize filter aids 4 understanding and diagrams with applications-plate and frame, rotary filters, filter aids-importance and applications text / reference t/r book title/author earle r l unit operations in food processing. web edition. pergamon press, r1 uk2004 fellows p j, food processing technology principles and practice, crc press, r2 2000. geankoplis c j, transport process and unit operations, prentice hall intl edition, r3 1993 albert ibarz and gustavo v. barbosa-cánovas, unit operations in food r4 engineering, crc press,2003 coulson, j.m. and etal. "coulson & richardson's chemical engineering", 6th r5 edition, vol. i & ii, butterworth – heinman / elsevier, 2004 foust, a.s. etal., " principles of unit operations", 2nd edition, john wiley & r6 sons, 1999. r. paul singh and dennis r. heldman, introduction to food engineering, 5th ed. r7 elsevier ,2014 online resources sl.no website link 1
<https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html> 2
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<https://savree.com/en/encyclopedia/ball-mill> 4
<https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html>
 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003

Aspect	What to prepare
	Coulson, J.M. and etal. "Coulson & Richardson's Chemical Engineering", 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., " Principles of Unit Operations", 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014 Online Resources Sl.No Website Link 1 https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html 2 https://www.sciencedirect.com/science/article/abs/pii/S0032591075850042 3 https://savree.com/en/encyclopedia/ball-mill 4 https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press,2003 Coulson, J.M. and etal. "Coulson & Richardson's Chemical Engineering", 6th R5 Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004 Foust, A.S. etal., " Principles of Unit Operations", 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed.

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<https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html>

2 <https://www.sciencedirect.com/science/article/abs/pii/S032591075850042>

3 <https://savree.com/en/encyclopedia/ball-mill> 4

<https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html>

മലയാളത്തിൽ നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം

നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical

terms English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of. (3 marks)**

► **Define or explain 7 Understanding Natural, forced circulation & falling thin film evaporator- Describe Single effect and multiple effect. (3 marks)**

► **Define or explain 4 Understanding evaporator evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators.. (3 marks)**

Module 2

► **Define or explain Describe the Principle of freezing 7 Understanding List the factors influencing freezing time. (3 marks)**

► **Define or explain 3 Remembering Describe the direct freezing systems. (3 marks)**

► **Define or explain 3 Understanding Describe the Indirect freezing systems. (3 marks)**

► **Define or explain 3 Understanding factors influencing freezing time, Freezing systems; Direct contact systems (air blast, immersion) &**

indirect contact systems. (3 marks)

Module 3

► **Define or explain laws 5 Understanding Discuss the equipment of size reduction in solid. (3 marks)**

► **Define or explain food 5 Understanding Discuss the equipment of size reduction in liquid. (3 marks)**

► **Define or explain 3 food Understanding mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill Describe the theory, importance, equipment, mechanisms, and. (3 marks)**

Module 4

► **Define or explain importance of filtration. (3 marks)**

► **Define or explain Discuss about the filtration equipment. (3 marks)**

► **Define or explain Summarize filter aids 4 Understanding and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications Text / Reference T/R Book Title/Author Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, R1 UK2004 Fellows P J, Food Processing Technology Principles and Practice, CRC Press, R2 2000. Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, R3 1993 Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food R4 Engineering, CRC Press, 2003 Coulson, J.M. and etal. "Coulson & Richardson's Chemical Engineering", 6th R5 Edition, Vol. I & II, Butterworth - Heinman / Elsevier, 2004 Foust, A.S. etal., " Principles of Unit Operations", 2nd Edition, John Wiley & R6 Sons, 1999. R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. R7 Elsevier ,2014**

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<https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html> 2

<https://www.sciencedirect.com/science/article/abs/pii/S0032591075850042>

3 <https://savree.com/en/encyclopedia/ball-mill> 4

<https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html>. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Discuss the Principle of evaporation 6 Understanding Explain the working principle and mechanism of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe the Principle of freezing 7 Understanding List the factors influencing freezing time.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **laws 5 Understanding Discuss the equipment of size reduction in solid.**

Module 4

1-mark definition: State the meaning of **importance of filtration.**

3-mark explanation: Explain one principle, process or comparison from this module.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html>
- <https://www.sciencedirect.com/science/article/abs/pii/S0032591075850042>
<https://savree.com/en/encyclopedia/ball-mill>
- <https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5421. Verify institutional updates against the current official curriculum.